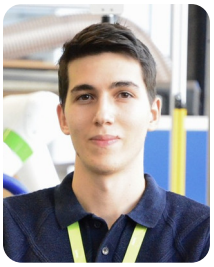
Suavi Yildirim

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Summary

Mechanical Engineer and Master's candidate with 5+ years of experience in robotics and automation, in the final semester of the M.Sc. Autonomy Technologies at FAU.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg — Erlangen, DE
Master of Science, Autonomy Technologies — 04/2024 – Expected 03/2027

- Coursework completed; Master's thesis at Institute FAPS written, evaluation pending
- Thesis: Pneumatic levitation of wires using CFD simulation and machine-learning regression

Ozyegin University — Istanbul, TR
Bachelor of Science, Mechanical Engineering — 09/2017 – 07/2022

- 100% Merit Scholarship, Honor Student
- Grade 1.8 (German scale); graduated 1st in Mechanical Engineering

Experience

Institute for Factory Automation and Production Systems (FAPS) — 07/2024 – Present
Graduate Research Assistant (HiWi) — Nürnberg, DE

- Coordinated development and validation of an automated pneumatic wire transport system
- Integrated Siemens S7-1500 PLCs, HMI panels, and sensors in TIA Portal; tuned control parameters

Cobro Robot Technologies — 09/2023 – 05/2024
Concept Design Engineer – PreSales Robotics Automation — Istanbul, TR

- Translated customer requirements into robotic automation concepts for industrial B2B applications
- Developed a flexible feeder system for vision-guided handling of randomly oriented parts

Arche Robotics — 07/2020 – 08/2023
Mechanical Design Engineer – Autonomous Robotics — Istanbul, TR

- Technical team lead for autonomous mobile robots, from design to commissioning at customer sites
- Delivered two fully operational AMR systems in MediaMarkt stores

Projects & Awards

Siemens Healthineers Robotics & AI Hackathon | *Teach the Robot (VLA)* — 10/2026

- Robotics & simulation engineer (team of 6): simulation-based robot-learning loop for a bimanual Franka FR3 handover in Isaac Sim and Genesis; **12× faster** demo generation

Siemens x Sony Immersive Design Challenge | *BatteryTwin-XR* — 12/2024 – 04/2025

- Led Team NextCycle (XR digital twin for EV batteries); **1st place** of 900 participants

OzU Rover Team | *Ozyegin University* — 09/2017 – 04/2020

- Team lead (2018–2020); 13th of 56 teams at ERC 2019; two conference publications

Volunteering & Extracurricular Activities

- **Siemens Campus Connect Student Ambassador** (since 05/2025); **Board Member**, FAU Space Gene
- **Panelist and speaker** on engineering and competitions; event support at Institute FAPS

Technical Skills

CAD & Simulation: Siemens NX, CATIA V5, SolidWorks, Fusion, CFD, Isaac Sim
Automation: Siemens S7-1500 / TIA Portal, FANUC robots
Programming: Python, C++, MATLAB, OpenCV, ROS
Manufacturing: 3D printing (FDM, SLA), CNC basics, electronics prototyping
Languages: English (professional working proficiency), German (B2.1)

Neumarkt i.d.OPf., October 30, 2026 — Suavi Yildirim

SUAVI YILDIRIM

M.Sc. AUTONOMY TECHNOLOGIES, FAU ERLANGEN-NÜRNBERG
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PORTFOLIO

Selected engineering projects in robotics, automation and product design — from mechanical design to simulation and robot learning.

Online portfolio with videos and more pictures: **suavi-yildirim.github.io/portfolio/**

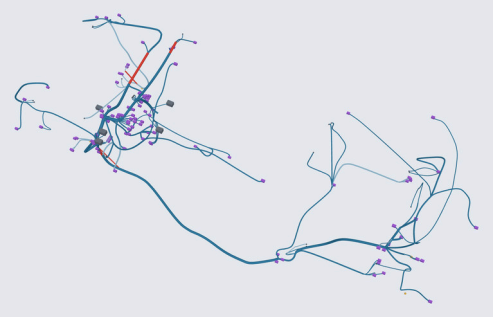
PROJECT	ORGANISATION / PERIOD	HIGHLIGHT	PAGE
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FORMBOARD XR	FAU – Institute FAPS 10/2026	KBL data in VR: wire harness assembly trainer, live in the browser	3
PNEUMATIC WIRE TRANSPORT (WTE)	FAU – Institute FAPS since 07/2024	Stable wire transport at 20 m/s	4
WIRE FLOATING EXPERIMENT (WFE)	FAU – Institute FAPS 2025 – 2026	Closed-loop wire levitation test rig	4
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AT1 MOBILE PLATFORM	Arche Robotics 2020 – 2023	AMR platform for 50 kg payload	10
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SURGICAL ROBOT FOR AUTONOMOUS SUTURING	OzU Robotics Lab	TÜBİTAK 1001 research project	12
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THE HOPE	OzU Rover Team 2019 – 2020	Top 36 of 102 teams (URC 2020)	13
THEIA	OzU Rover Team 2018 – 2019	13th of 56 teams (ERC 2019)	13

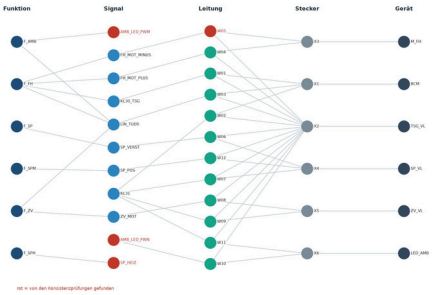
WIRE HARNESS KNOWLEDGE GRAPH

FAPS Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
Prof. Dr.-Ing. Jörg Franke

FAU – Institute FAPS · 10/2026 · Developer (graduate research assistant)

KBL · RDF/OWL · SPARQL · Python · Three.js





What?

- **KBL** (VDA 4964) describes the wire harness: connectors, wires, modules. Functions and control units live in the **E/E model**. Both only share **names**.
- Goal: link both views in **one semantic graph** and answer questions that neither source can answer alone.
- Developed to understand the **KBL data formats** in depth, from the XML structure to the 3D geometry.

How?

- Small **ontology** (OWL, 12 classes, labels in German and English, naming conventions) and a Python converter (rdflib) for KBL and E/E data.
- **SPARQL** questions for traceability, change impact, variants and consistency; first on a fictitious door harness, then on the public **VW "Old Beetle"** sample harness (KBL 2.4).
- **3D routing** rebuilt from the KBL B-spline centre lines; interactive viewer (Three.js).

Results

- **5 957 triples**; variants compared: MAXI 254 connections and 278 m of wire, MINI 165 and 218 m.
- Consistency checks found an **empty module** and **5 segments** whose geometry does not match the topology.
- Live demo (EN/DE): suavi-yildirim.github.io/bordnetz-knowledge-graph

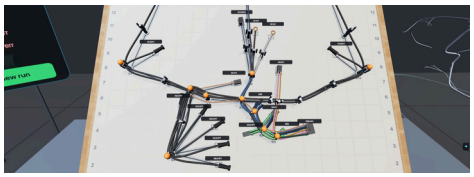
FORMBOARD XR

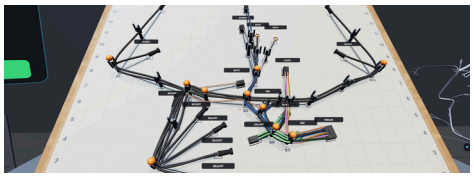
FAPS Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
Prof. Dr.-Ing. Jörg Franke

FAU – Institute FAPS · 10/2026 · Developer (graduate research assistant)

WebXR · Meta Quest 3 · KBL · Three.js · Python







What?

- Harness assembly is manual work on a **formboard**. An XR assistance system needs the harness as a **1:1 board layout**, which the KBL file does not contain.
- Goal: a **data-driven** trainer that runs on a Meta Quest 3 without installation.

How?

- Python pipeline: the **KBL** (VDA 4964) of the public VW "Old Beetle" is flattened to a formboard. Segments keep their true length; pegs are derived from the topology.
- **WebXR** app (Three.js) with hand tracking: route wire by wire, every peg is checked; guided and test mode; time, errors and hints are logged. Connectors and colours come from the KBL.

Results

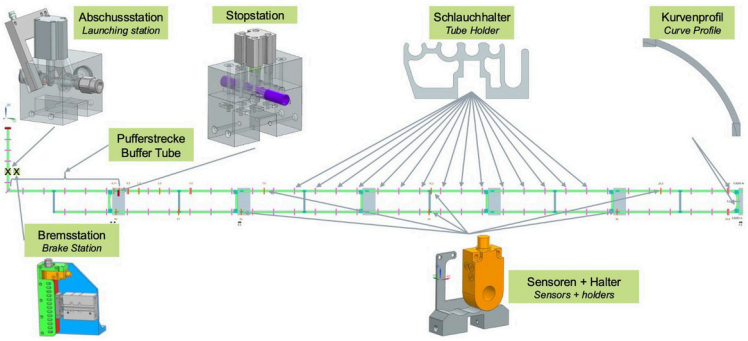
- **18 wires, 20 connectors**; the KBL wire lengths match the board paths to the millimetre.
- Live demo: suavi-yildirim.github.io/formboard-xr

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PNEUMATIC WIRE TRANSPORT (WTE)

FAU – Institute FAPS · since 07/2024 · Graduate research assistant

Siemens NX | 3D printing | Pneumatics





What?

- As part of **Project SIMON** at the **Institute FAPS**, I contributed to an **automated pneumatic "shooting" system** that replaces manual **wire handling** and removes assembly bottlenecks for flexible wire components.

How?

- Used **Siemens NX** for CAD modeling of the **experimental setup**, including tube holder, stop and brake stations.
- Used **FDM and SLA 3D printing** for parts with different **structural** and high-tolerance **aerodynamic** requirements.

Results

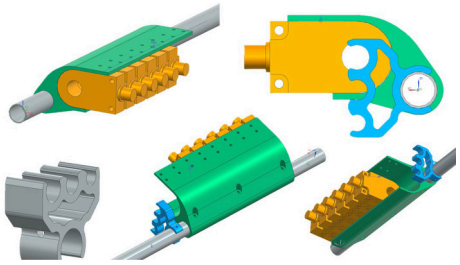
- **Achieved stable transport** at **20 m/s** while keeping a defined state of order for flexible leads.
- **Quantified** empirical pressure-loss coefficients for industrial-scale pneumatic system modeling.

WIRE FLOATING EXPERIMENT (WFE)

FAU – Institute FAPS · 2025 – 2026 · Graduate research assistant, Master's thesis

S7-1500 | TIA Portal | PID | CFD







What?

- Vertical test rig in which an **upward airflow holds a wire** in floating equilibrium – the basic case of every lifting section in **pneumatic wire transport**.

How?

- Designed launching station, tube and **sensor holders** (Siemens NX, 3D printing).
- **Control cabinet**: S7-1500 software PLC, ET 200SP, IO-Link, Festo proportional pressure regulator and flow sensor.
- **PID control** of the inlet pressure from 5 inductive sensors; HMI for automated test runs.

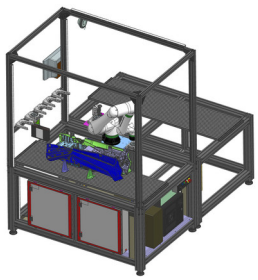
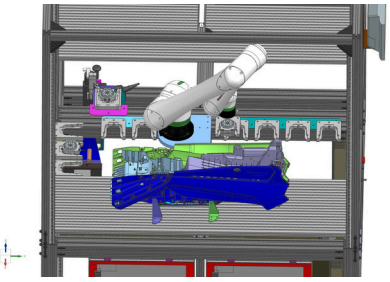
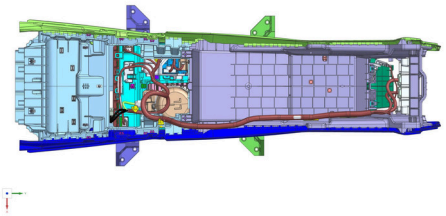
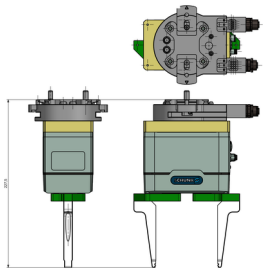
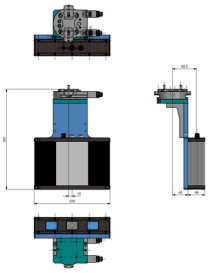
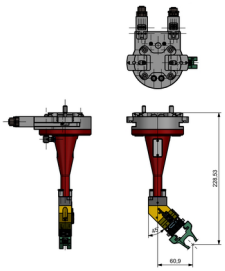
Results

- **Reproducible** floating tests for wires of **0.5–3.5 m** and **0.5–6 mm²**.
- Dataset for the Master's thesis: **CFD** validation and **machine-learning** prediction of floating conditions.

NEXT2OEM – AUTOMATED WIRE HARNESS PRODUCTION

FAU – Institute FAPS × Audi · 2024 – 2026 · Graduate research assistant

Wire harness | Siemens NX | 3D printing

What?

- **NEXT2OEM** is an **Audi-led** research project (BMWK, 2023–2026) with FAPS and industry partners.
- Goal: **digitise and automate** the whole wire-harness value chain, from connector to vehicle assembly, to bring harness production back to Germany.

How?

- Helped **build the demonstrator**: fixtures, electric and pneumatic **gripper fingers** and a **camera mount** for the robot stations (Siemens NX).
- **Redesigned Audi parts** for 3D printing (FDM, PETG/PLA).

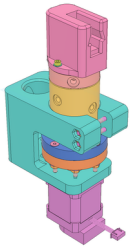
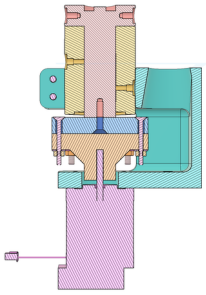
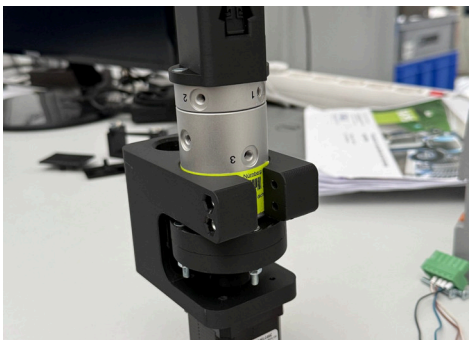
Results

- Gripper and camera tooling **integrated** in the demonstrator stations.
- **Print-ready** Audi parts for fast iterations of the setup.
- Project result: Audi validated the prototype on a centre-console harness in Ingolstadt.

WIRE TWISTING UNIT

FAU – Institute FAPS · Design and build

Wire harness | Stepper drive | Arduino | 3D printing

What?

- A compact **twisting unit** for the **Robotik Challenge** setup: it holds the wire end in a **connector holder** and turns it to **twist the wire**.

How?

- Designed the **bracket, motor coupling and connector holder** around an igus stepper motor and a rotary feedthrough.
- **Closed-loop stepper** drive (CL42T) with an **Arduino Nano**; speed tests up to 600 rpm.
- Printed parts for fast iterations; mounts directly on the task board.

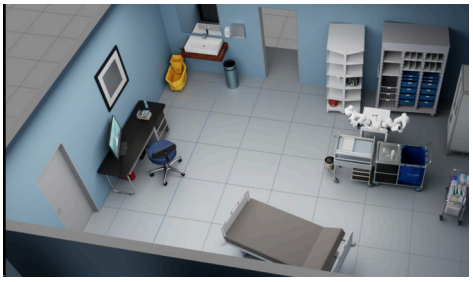
Results

- **Built and tested** unit, integrated in the Robotik Challenge task board.
- Makes **wire twisting** – a typical wire-harness process step – available as a robot task.

ROBOTICS & AI HACKATHON – TEACH THE ROBOT

Siemens Healthineers · 10/2026 · Robotics & Simulation Engineer, team of 6

Isaac Sim | Genesis | LeRobot | VLA | Quest 3







What?

- 3-day **Siemens Healthineers Robotics & AI Hackathon 2026** in Erlangen: build the **learning loop** for a **vision-language-action (VLA)** robot companion in the cathlab.
- Task: bimanual box handover with a **Franka FR3 Duo** in simulation.

How?

- **Robotics & Simulation Engineer** in a 6-person team.
- Built a **scripted IK expert** in **NVIDIA Isaac Sim 6.0** that generates demos with moved box poses (16/18 successful).
- Fine-tuned $\pi_{0.5}$ **(LoRA)** and **ACT**; rebuilt the task in **Genesis** for parallel data generation.

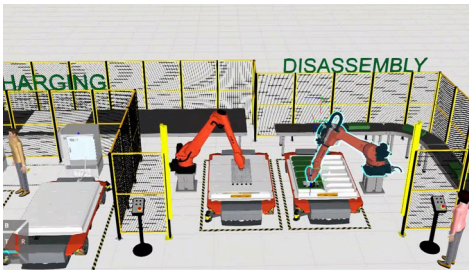
Results

- Showed that the **data, not the model**, was the bottleneck: the 94 provided demos contain one single motion.
- **12× faster** demo generation: from 6 min to about 30 s per demo.
- Prototyped **VR teleoperation with Meta Quest 3** for human demos.

IMMERSIVE DESIGN CHALLENGE

Siemens × Sony · 12/2024 – 04/2025 · Team lead

XR | Digital Twin | NX









What?

- The Immersive Design Challenge is a **global engineering competition** organized by Siemens Digital Industries Software in collaboration with Sony.
- Encourages **sustainable design** by bringing immersive technologies such as **XR and digital twins** into real engineering problems.
- Scale: ~**900 students**, **230+ universities**, **38 countries**.

How?

- **Led** the team **NextCycle**.
- **Developed** BatteryTwin-XR: a Siemens-integrated **digital twin** for a sustainable **EV battery lifecycle**.
- **Designed** an XR framework with the Sony XR head-mounted display to support **battery monitoring, reuse and recycling** decisions.
- Focused on **circular economy**: better recycling efficiency, less battery waste.

Results

- **Won 1st place** among the international finalist teams.
- Recognized by Siemens and Sony for an **innovative, sustainability-driven** XR solution.
- Awarded a **Sony XR headset** and a 1-year **NX Immersive Designer** license for academic and research use.
- Invited to **Siemens Realize LIVE**; helped expand immersive design capabilities at FAU.

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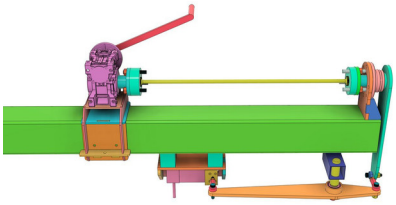
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suavi-yildirim.github.io/portfolio/

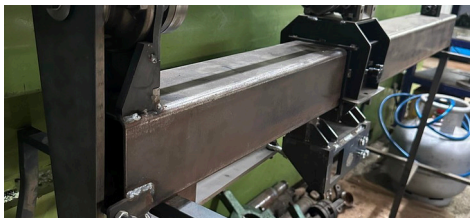
TORSIONAL MEASUREMENT DEVICE

2E Design · 2023 – 2024 · Mechanical design

Test rig | Load cell | Motorsport







What?

- **Developed** a **torsional measurement** device for analysing the mechanical characteristics of **anti-roll bars** in automobiles.

How?

- **Implemented** a hub and bearing setup to assess **anti-roll bar twisting**.
- **Transferred** the load through lever arms to a **1-ton capacity** load cell for accurate measurement.

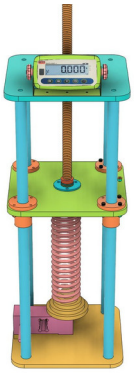
Results

- The device allowed **measurement and adjustment** of anti-roll bar twist **stiffness**, ensuring optimal performance for the targeted **racing** vehicles.

SPRING STIFFNESS MEASUREMENT DEVICE

2E Design · 2023 – 2024 · Mechanical design

Test rig | Motorsport



What?

- **Developed** a **spring stiffness measurement** device for analysing springs used in racing **car suspension systems**.

Results

- The device enabled accurate measurement of **spring coefficients**, allowing the suspension to be tuned for **optimal track performance**.

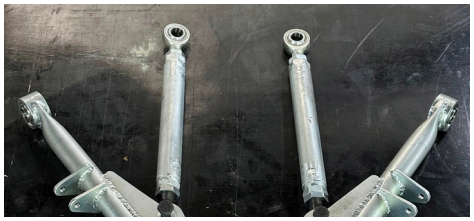


SUSPENSION CONTROL ARM OF A RALLY CAR

2E Design · 2023 – 2024 · Mechanical design

Redesign | Manufacturing | Motorsport







What?

- Conducted extensive **R&D** to improve the **suspension control arm** of a **rallycross** vehicle, to optimise performance and durability under demanding racing conditions.

How?

- **Redesigned** the suspension arm after the **original broke** during a race.
- **Manufactured** a more **rigid and robust** arm and integrated it into the vehicle.

Results

- The new design gave a **stronger, more durable** suspension arm, clearly improving **vehicle performance** and reliability on the track.

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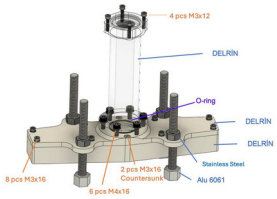
suaviyildirim.github.io/portfolio/

GRIPPER DESIGN FOR FOOD PACKAGING

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

ABB Delta | Vacuum gripping







What?

- **Developed** robot grippers for **ABB Delta robots**, tailored to the operational and hygiene requirements of a high-speed **food production company**.

How?

- Designed grippers for high-speed **pick-and-place** operations.
- **Optimised gripper mass** to meet the robot payload limits.
- **Used O-ring sealing** for reliable, repeatable suction performance.

Results

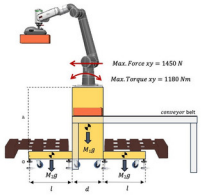
- **Integrated** into the production line: **20% higher** packaging throughput.
- **90% cost reduction** by replacing standard components with custom suction cups.

COBOT PALLETIZER

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

ABB CRB 15000 | Cobot | Structural analysis







What?

- **Developed** a **mobile palletizing robot** with collaborative robots that moves easily between **production lines** for more flexibility and efficiency.

How?

- Performed **static and dynamic analyses** to design a robust base fixture.
- Selected the **ABB CRB 15000** cobot with a **12 kg payload**.

Results

- The system was **successful** and helped the company win further projects, as it stands out from stationary palletizers on the market.

FLEXIBLE FEEDER SYSTEM

Cobro Robot Technologies · 2023 – 2024 · Concept design engineer

Machine vision | Bin picking







What?

- A **flexible feeding system** for manufacturers that need robotic automation to **separate mixed products**.

How?

- A camera above the feeder locates **randomly oriented parts**; the robot picks them (**vision-guided handling**).

Results

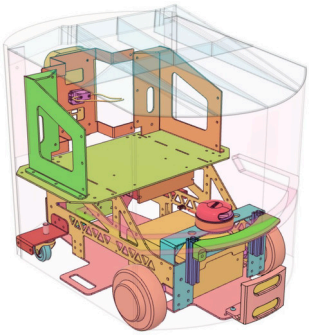
- A **cost-effective** automated feeder that sorts mixed parts, ready to be extended with **image processing and AI**.

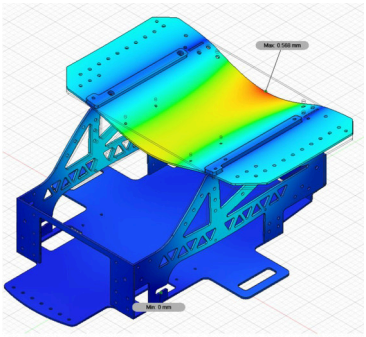
AT1 MOBILE PLATFORM

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

SolidWorks | Fusion 360 | FEA







What?

- **AT1** is an indoor **autonomous mobile robot** platform (AMR).
- A versatile platform for **multiple applications**.
- Designed the chassis for a **50 kg** payload.

How?

- Used **SolidWorks** and **Fusion 360** for the mechanical design.
- Ran an **FEA study** of the forces and moments on the chassis.
- 3 mm **6061 aluminium** sheet-metal parts joined by TIG welding.

Results

- **Launched** AT1, now used from **healthcare** to **retail** and **hospitality**.
- Basis for ongoing development of new hardware and software modules.

PROMOBOT

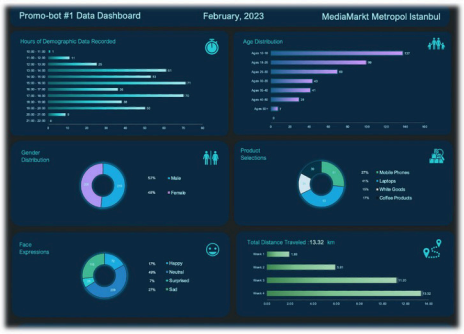
ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

AMR | Retail | Dashboards









What?

- Promobot helps businesses **engage customers** and build their brand in physical and digital stores.
- Features: **welcoming**, **customer guidance**, **marketing** and **AR applications**.

How?

- A **data-driven** companion that supports **digital transformation**.
- Monthly dashboards of sensor data such as **air quality** and **demographics**.

Results

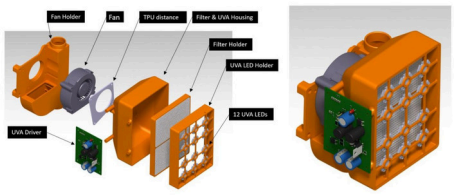
- **Deployed** at **MediaMarkt** Metropol Istanbul on 2 Jan 2023.
- Interacted with **400 customers** per month on average.
- Drove **10+ km** autonomously per month without mechanical issues.

PURA

ARCHE ROBOTICS

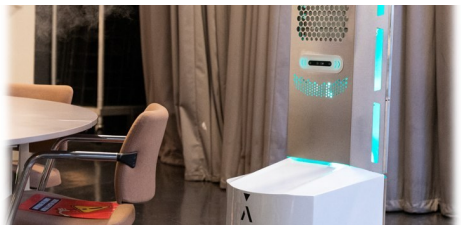
Arche Robotics · 2020 – 2023 · Mechanical design engineer

Photocatalysis | Air sensing









What?

- **PURA** is based on the AT1 platform and **sanitizes** surfaces and floors and **purifies the air**.
- Designed a sanitation chamber with a **photocatalyst module** and **HOCl disinfection** liquid.
- Senses **air quality**: particles, VOC, humidity, temperature.

How?

- Photocatalyst module with **16 ceramic foams** coated with TiO_2 and **40 UVA LEDs**.
- A **piezo fog** generator makes a **dry fog** of disinfectant, blown by a DC fan at **150 m^3/h**.
- No consumables such as HEPA filters, cartridges or mercury UVC bulbs.

Results

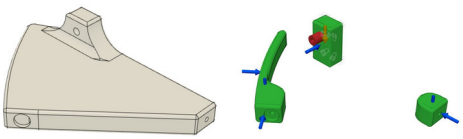
- **Deployed** an autonomous disinfection robot that works **24/7**.
- Cleans as much air per hour as **300 people** breathe.
- Works **without harmful chemicals**.
- Built strategic relationships with Chinese manufacturers.

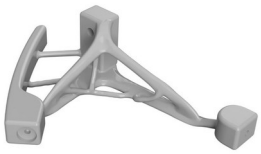
GENERATIVE DESIGN STUDY

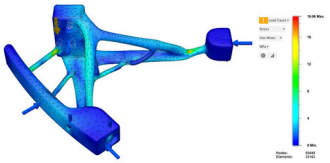
ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

Fusion 360 | Generative design







What?

- Used **generative design** (AI-driven algorithms) to create **lightweight parts** that meet performance and design requirements.

How?

- Used the **Fusion 360** generative design toolbox.
- Defined the **loads** on the part; compared **von Mises stress** results to ensure structural integrity.

Results

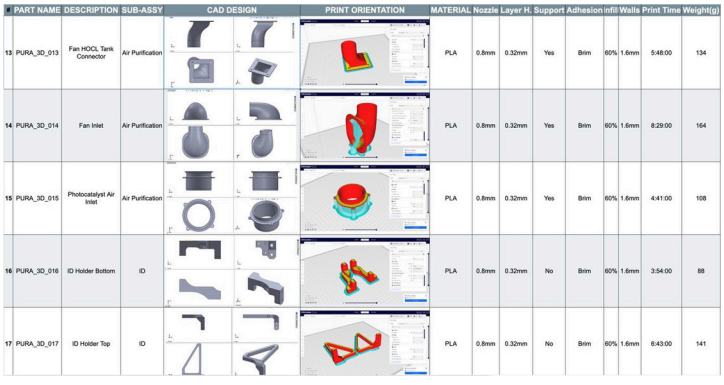
- **30% less weight** with all design constraints and performance criteria met.

ADDITIVE MANUFACTURING PART MANAGEMENT

ARCHE ROBOTICS

Arche Robotics · 2020 – 2023 · Mechanical design engineer

FDM | Part tracking



#	PART NAME	DESCRIPTION	SUB-ASSY	CAD DESIGN	PRINT ORIENTATION	MATERIAL	Nozzle	Layer H.	Support	Adhesion	infil	Walls	Print Time	Weight(g)
13	PURA_3D_013	Fan HOCL Tank Connector	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	5:48:00	134
14	PURA_3D_014	Fan Inlet	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	8:29:00	164
15	PURA_3D_015	Photocatalyst Air Inlet	Air Purification			PLA	0.8mm	0.32mm	Yes	Brim	60%	1.6mm	4:41:00	108
16	PURA_3D_016	ID Holder Bottom	ID			PLA	0.8mm	0.32mm	No	Brim	60%	1.6mm	3:54:00	88
17	PURA_3D_017	ID Holder Top	ID			PLA	0.8mm	0.32mm	No	Brim	60%	1.6mm	6:43:00	141

What?

- Created an **Excel-based tracking system** to document 3D-printed parts and keep an organized **inventory**.

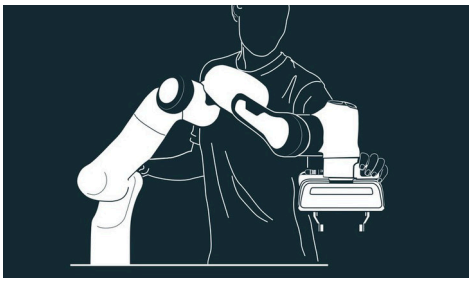
Results

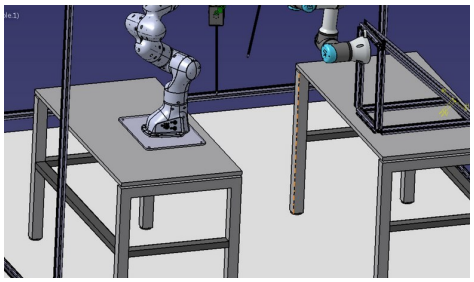
- **Standardized** part quality through identical slicing settings.
- **Efficient tracking** of series production, total print time and material use.

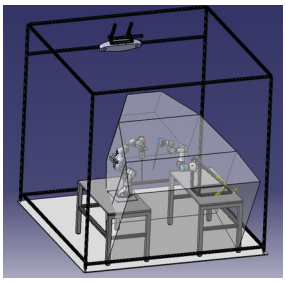
SURGICAL ROBOT FOR AUTONOMOUS SUTURING

OzU Robotics Lab · Mechanical design

CATIA V5 | UR3 | Franka Panda







What?

- A **surgical robot** with two cobots, **UR3** and **Franka Panda**. I designed the **end-effector flanges** for the Franka Panda arms and the structure for the **3D vision sensors**.

How?

- Modeled robots and sensors in **CATIA V5**.
- Adapted the **Da Vinci EndoWrist** as end-effector.
- **Built** tables from welded rigid profiles to reduce vibration.

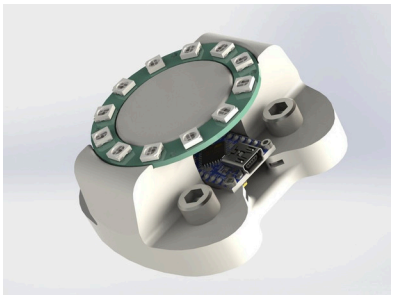
Results

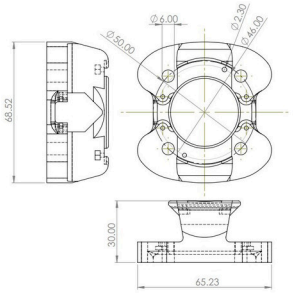
- Ongoing at the **OzU Robotics Laboratory**: "Development of Surgical Robot Assistants for Autonomous Suturing with Skill Transfer from Surgeon to Robot", funded by **TÜBİTAK 1001**.

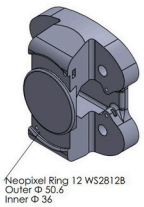
KUKA KR20-3 LIGHT PAINTING

Ozyegin University – RobArt · 2021 – 2022

KUKA | Rhino/Grasshopper







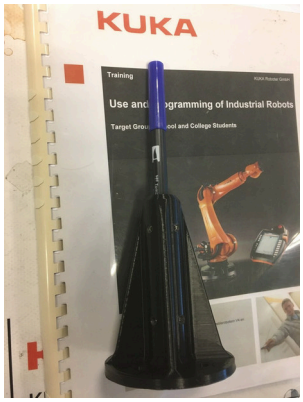
What?

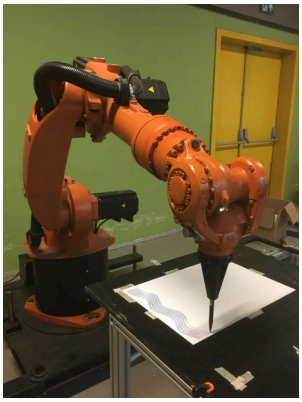
- In my senior year at Ozyegin University I took part in the **RobArt** workshops with industrial robot arms.
- **Light painting**: programmed a **KUKA KR20-3** with **Rhino/Grasshopper** and photographed the planned trajectory.
- **Designed and 3D printed** a flange holding an **LED ring**.

KUKA KR20-3 PEN PLOTTER

Ozyegin University – RobArt · 2021 – 2022

KUKA | 3D printing





What?

- A second RobArt workshop on **pen-plotter art**: the robot arm draws 2D images.
- **Designed and 3D printed** a flange holding a pen on the KUKA KR20-3.

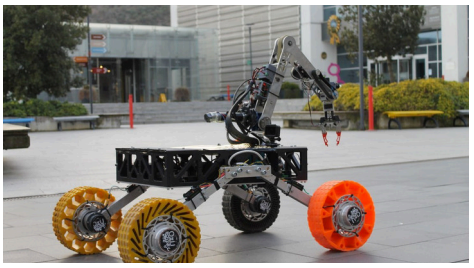
THE HOPE

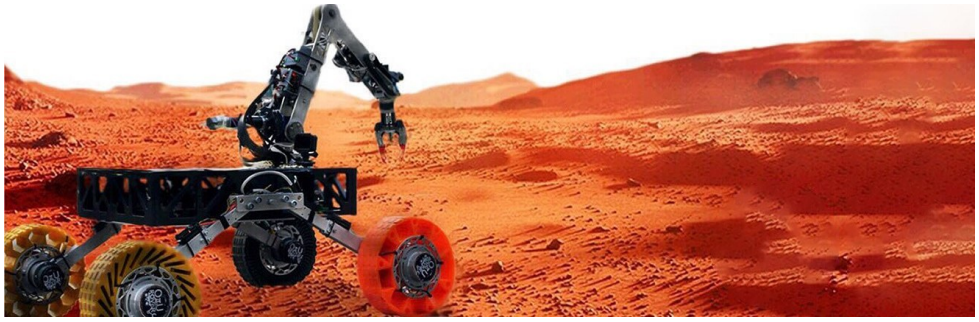
OzU Rover Team · 2019 – 2020 · Team lead

Mars rover URC









What?

- **The Hope** is a Mars rover prototype built by the OzU Rover Team for the **University Rover Challenge 2020** in the USA.

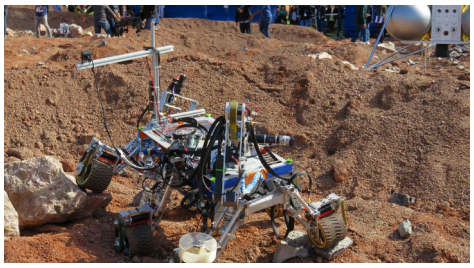
Results

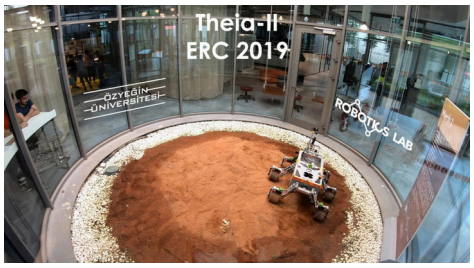
- **Qualified**, ranking in the top 36 of 102 teams.
- The event was cancelled due to COVID-19.

THEIA

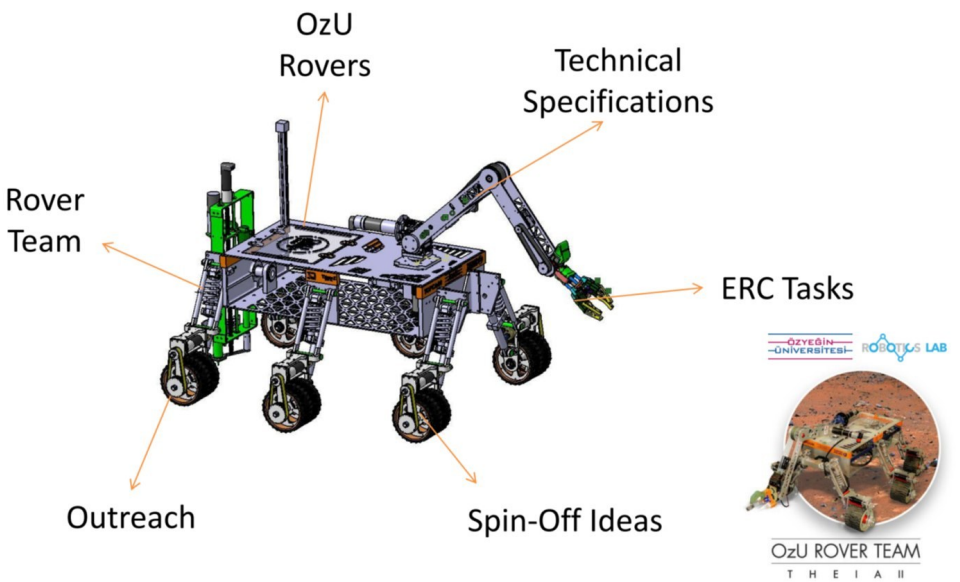
OzU Rover Team · 2018 – 2019 · Mechanical designer, then team lead

Mars rover ERC









What?

- Theia was built by the OzU Rover Team for the **European Rover Challenge**.
- Teams design and build a **Mars rover prototype** for **tele-operated** or **autonomous** tasks that assist astronauts.

Results

- **ERC 2018: 26th/65** as mechanical designer.
- **ERC 2019: 13th/56** as **team lead**, through leadership and design optimization.